

*Cognition, Communication and
Survey Measurement*
SurvMeth632

Fall, 2024
Aaron Maitland

8.28.24

SurvMeth 632

Fall, 2024

1. Course Description
2. Introductions
3. Thinking about the Response Process
4. Measurement Error

The Response Process

- Responding to a survey question requires more than retrieving pre-formed and accurate answers as if from a database
 - Info stored in respondents' (*R*'s) heads may not be available, complete, or accurate
 - *R* may reconstruct answer from fragments of remembered info and cues from environment and interviewer (if there is one)
- Scientific study of cognition can make evident the thinking that underlies response behaviors, and possibly reduce associated error
 - Language comprehension
 - Retrieval from autobiographical memory
 - Estimation, judgment, decision making

Measurement Error

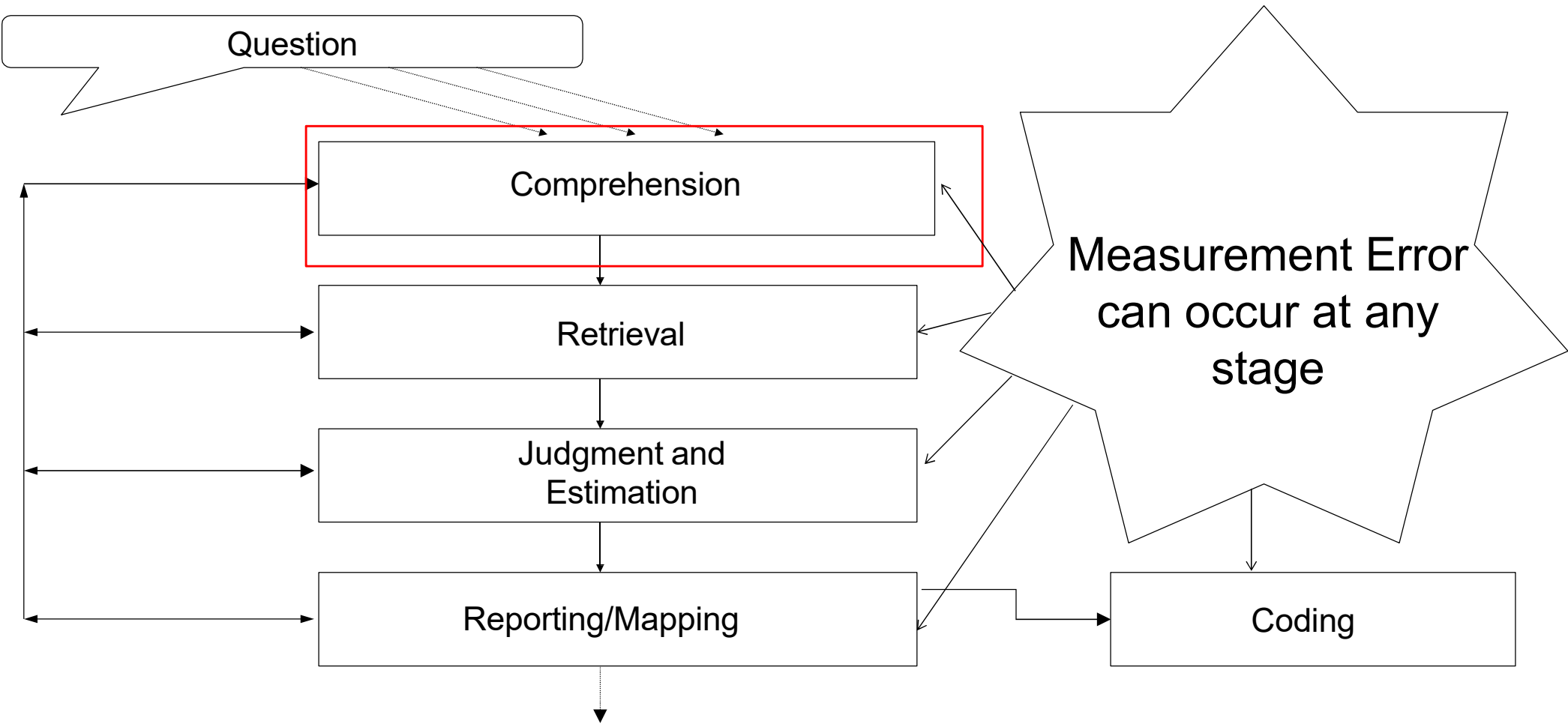
- This is a course in *measurement error*, i.e., the difference between what *Rs* report, i.e., what is measured, and the actual or “true” value
 - True value $\approx$ “ground truth” or “gold standard” – is an objective measure that is conventionally agreed to be the truth, e.g., health records or step count from smartphone
 - True value is sometimes thought about as the average of an infinite number of measurements
- Especially the psychology behind measurement error
- Not a questionnaire design course but can provide theoretical background for SurvMeth630
 - which is a questionnaire design course
- It’s a course about the interesting measurement problems that arise from how people think

Thinking about the response process

- “Models” of the response process
- Comprehension
- Retrieval
- Estimation and Judgment
- Response Mapping
- Classification/Coding (for open-ended responses)

“Models” of the response process

- Answering a survey question necessarily involves certain types of mental processes, generally considered to be ordered in stages
- There is broad agreement that the stages include:
 - Comprehension
 - Retrieval
 - Estimation and Judgment
 - Reporting (response formatting, response mapping)
- Each stage must begin before the next one begins
 - How would one know what to retrieve without interpreting the question?
- But some simultaneous processing is possible
 - e.g., retrieval of events based on one already-interpreted word may begin while next word is being interpreted
- And respondents may return to earlier stages (“backtracking”)
 - *R* may re-interpret *Q* if cannot find an appropriate response category



Comprehension Processes

- Comprehension usually assumed (by psychologists) to involve analysis at various levels:
 - sensory: speech stream are segmented into words, printed characters are recognized
 - lexical: retrieve word meaning(s), pronunciation, part of speech, etc.
 - syntactic: string words together into a grammatical sequence
 - semantic: determine overall sentence (question) meaning
 - pragmatic: infer speakers' intentions
- Comprehension problems can occur at each of these levels
- We will focus on semantic and pragmatic levels

Comprehension Processes:

Pragmatic inferences

- Grice's cooperative principle (CP) allows listeners* to infer meaning by assuming speaker** is *cooperating*:
 - Component theories (“maxims”):
 1. Quantity: say as much but no more than necessary
 2. Quality: be truthful
 3. Relation: make your contribution relevant
 4. Manner: be clear; avoid obscurity and ambiguity

Rs* in our case; *lwers* in our case

Pragmatic Inferences and Measurement Error

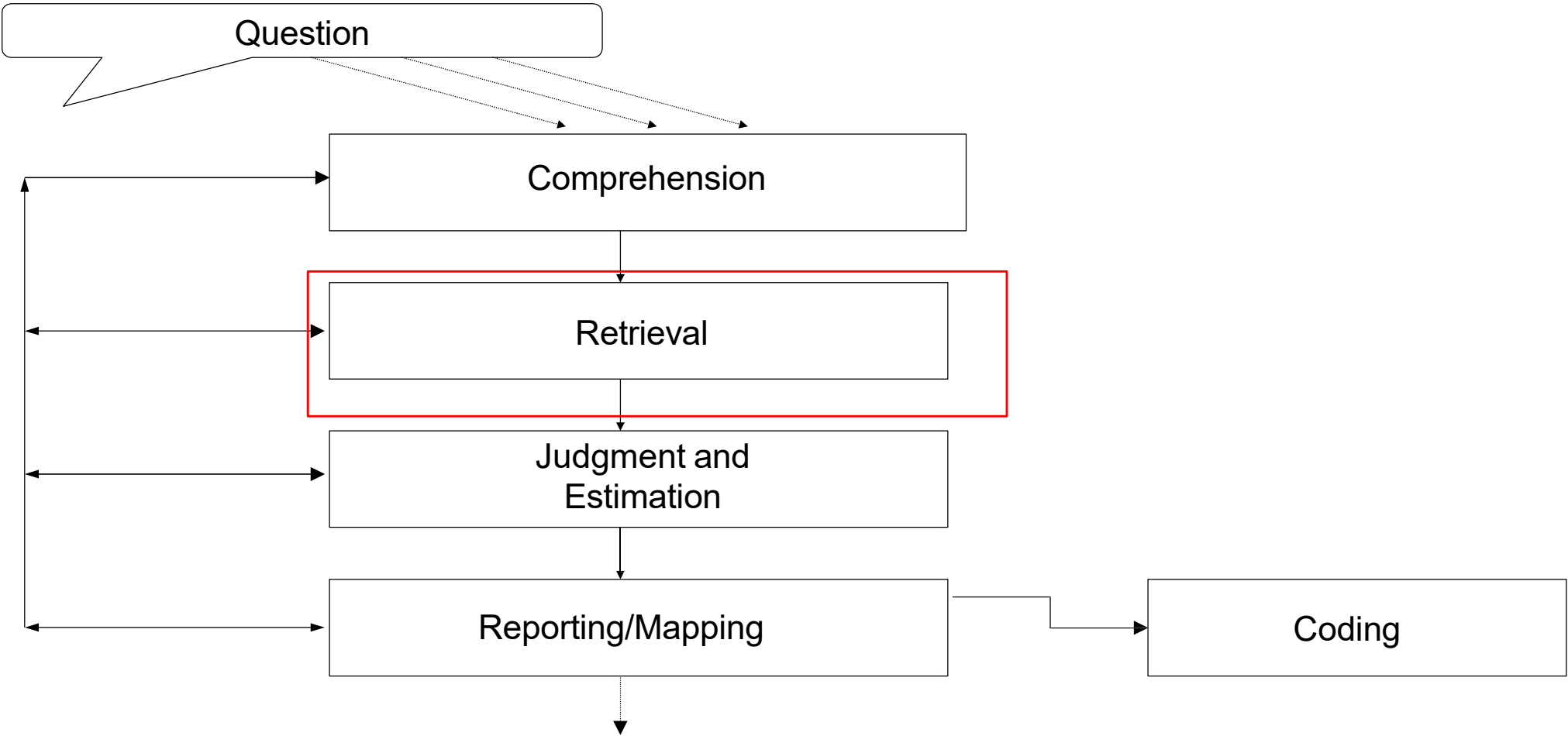
- Based on CP, *R* might reason, e.g., that interviewer or researcher
 - would not say something if he/she did not intend to inform me (*maxim of quantity*)
 - So, I will include that information in my interpretation of *Q*
- Impact of Gricean inference on measurement error
 - *Rs* infer interviewer/researcher “intentions” that were not actually intended, leading to misinterpretation of *Q*

Example: Gricean Inference Can Lead to Misinterpretation

- *R* may consider all available information to infer meaning:
 - Schwarz et al. (1991) asked *Rs* “How successful would you say you have been in life?”
 - *Rs* rated on 11-point scale with verbal end labels “not at all successful” and “extremely successful”
 - scale from 0 to 10 for one group, -5 to +5 for other
 - formally equivalent but numerical labels associated with verbal labels differ between groups
 - 34% of 0 to 10 group responded in lower half (0 to 5)
 - 13% of -5 to +5 group responded in lower half (-5 to 0)
 - Authors propose that 0 is combined with “not at all successful” to mean “absence of success” and -5 is combined with label to mean “presence of failure”
- It’s unlikely designers intend for *Rs* to interpret scale values this way
- Unwarranted inference based on everyday conversational practices

Comprehension Processes: Interviewer-Respondent Interaction

- Respondents may exhibit cues of uncertainty that predict comprehension difficulty
 - Schober & Bloom (2004) observed *Rs*' answers contain more *ums* and *uhs* when question is complicated than straightforward
 - I: Did you have any purchases or expenses for household furniture?
 - R: um .. is a floor lamp 'household furniture'?
 - Question is complicated because 'household furniture' can mean many things
- If *R* unable to get clarification about meaning of household furniture, risk of misunderstanding is increased

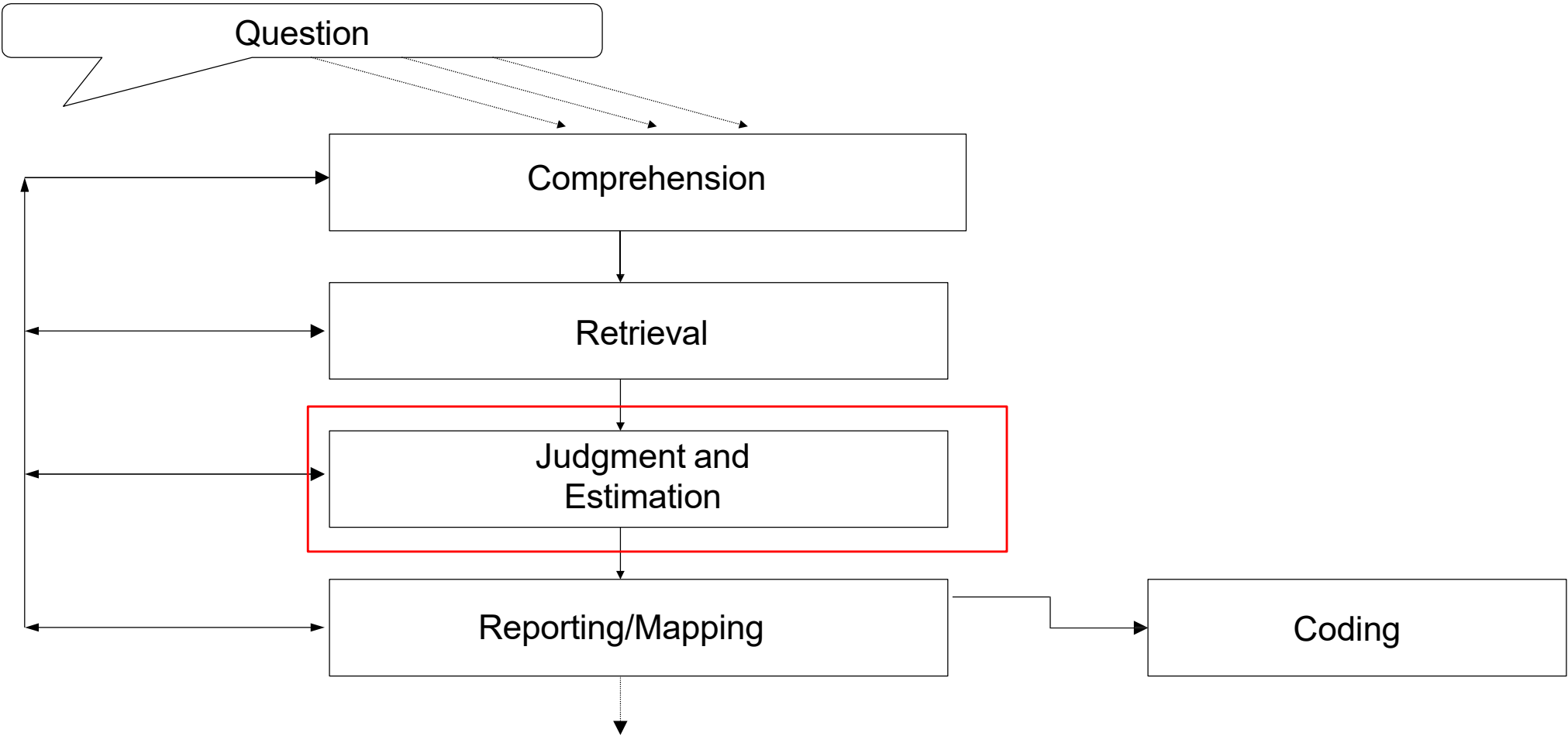


Recalling episodes vs. rates

- Menon (1993) asked *Rs* to judge the frequency of certain mundane activities, e.g., snacking, brushing one's teeth, drinking from a water fountain
- Activities occurred on either *regular* or *irregular* schedules and were either *similar* or *distinctive* each time they occurred
- Distinguished between responses based on episodic recall (remembered events) and rates (knowledge of temporal patterns of occurrence)
 - Assumed responses based on one strategy or the other
- Predicted most use of rates and, therefore, least use of recall when events are similar and regular; least use of rates and, therefore, most use of recall when events are distinctive (dissimilar) and irregular

	Similar	Dissimilar
Regular	89.5	62.5
Irregular	44.7	24.1

% Responses based on Rates

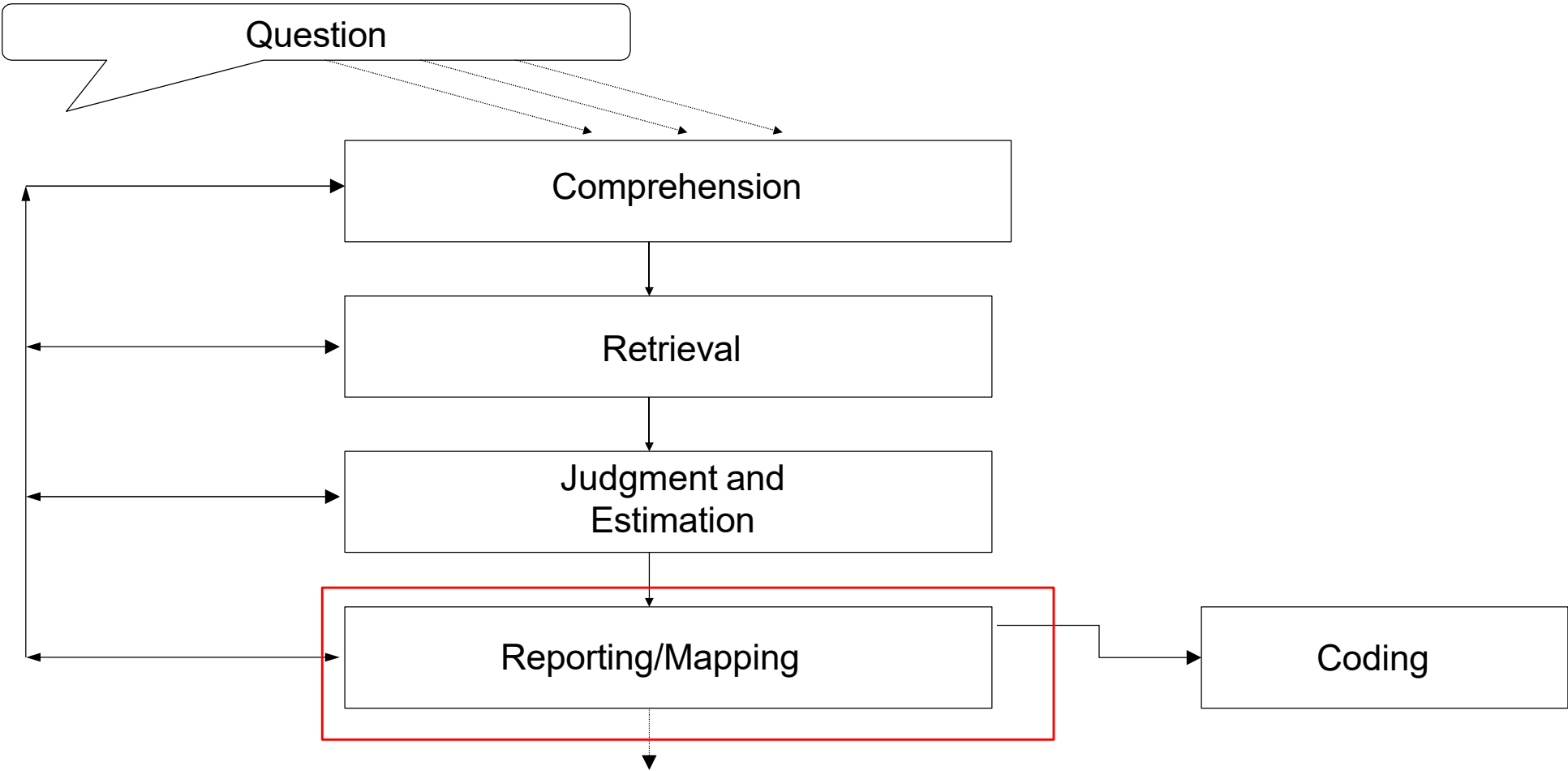


Estimation and Judgment

- Compensating for imprecise memory
 - when it is hard to recall relevant events and no rates come to mind, *Rs* can infer frequency on basis of retrieval difficulty -- *availability heuristic* (Tversky and Kahneman)
 - “I can’t recall many instances of the event so it must be rare”

Estimation and Judgment (2)

- Inferring speakers' intentions when answering attitude questions
 - Schwarz, et al. (1991) asked *Rs* a specific question (“How satisfied are you with your marriage?”) before a general one (“How satisfied are you with your life in general?”)
 - Answers to both questions correlated (*assimilation, priming, activation*)
 - unless *R* made aware of potential redundancy between *Qs*
 - in which case *R* reasons researchers would not ask same *Q* twice
 - So “subtracted out” their marriage in evaluating their life in general, reducing correlations between answers



Response Mapping/Selection

- Even after comprehension, retrieval, estimation and judgment are complete, *Rs* must map the results to available response options
- Response order
 - when response options are read out loud (usually by an *Iwer*), people show preference for the last one read (a “recency” effect)
 - when response options are self-administered, i.e., typically presented textually, people show a preference for the first one read (a “primacy” effect)
- Erroneously treating response categories as response distribution
 - Schwarz, Hippler, Deutsch and Strack (1985) asked *Rs* how many hours of TV they watch weekly by selecting frequency category from one of two response scales
 - Low frequency (middle: 1.5) and high frequency (middle: 3.5) scales

Scales in Schwarz et al. (1985)

Low frequency alternatives	%	High frequency alternatives	%
Up to $\frac{1}{2}$		Up to $2\frac{1}{2}$	
$\frac{1}{2}$ to 1		$2\frac{1}{2}$ to 3	
1 to $1\frac{1}{2}$		3 to $3\frac{1}{2}$	
$1\frac{1}{2}$ to 2		$3\frac{1}{2}$ to 4	
2 to $2\frac{1}{2}$		4 to $4\frac{1}{2}$	
$> 2\frac{1}{2}$		$> 4\frac{1}{2}$	

Weekly hours of TV watching

Scales in Schwarz et al. (1985)

Low frequency alternatives	%	High frequency alternatives	%
Up to ½	7.4	Up to 2½	62.5
½ to 1	17.7	2½ to 3	23.4
1 to 1 ½	26.5	3 to 3 ½	7.8
1 ½ to 2	14.7	3 ½ to 4	4.7
2 to 2 ½	17.7	4 to 4 ½	1.6
> 2 ½	16.2	> 4 ½	0.0

Weekly hours of TV watching

For low frequency scales, 16.2% reported > 2.5 hours

For high frequency scales, 37.5% reported > 2.5 hours

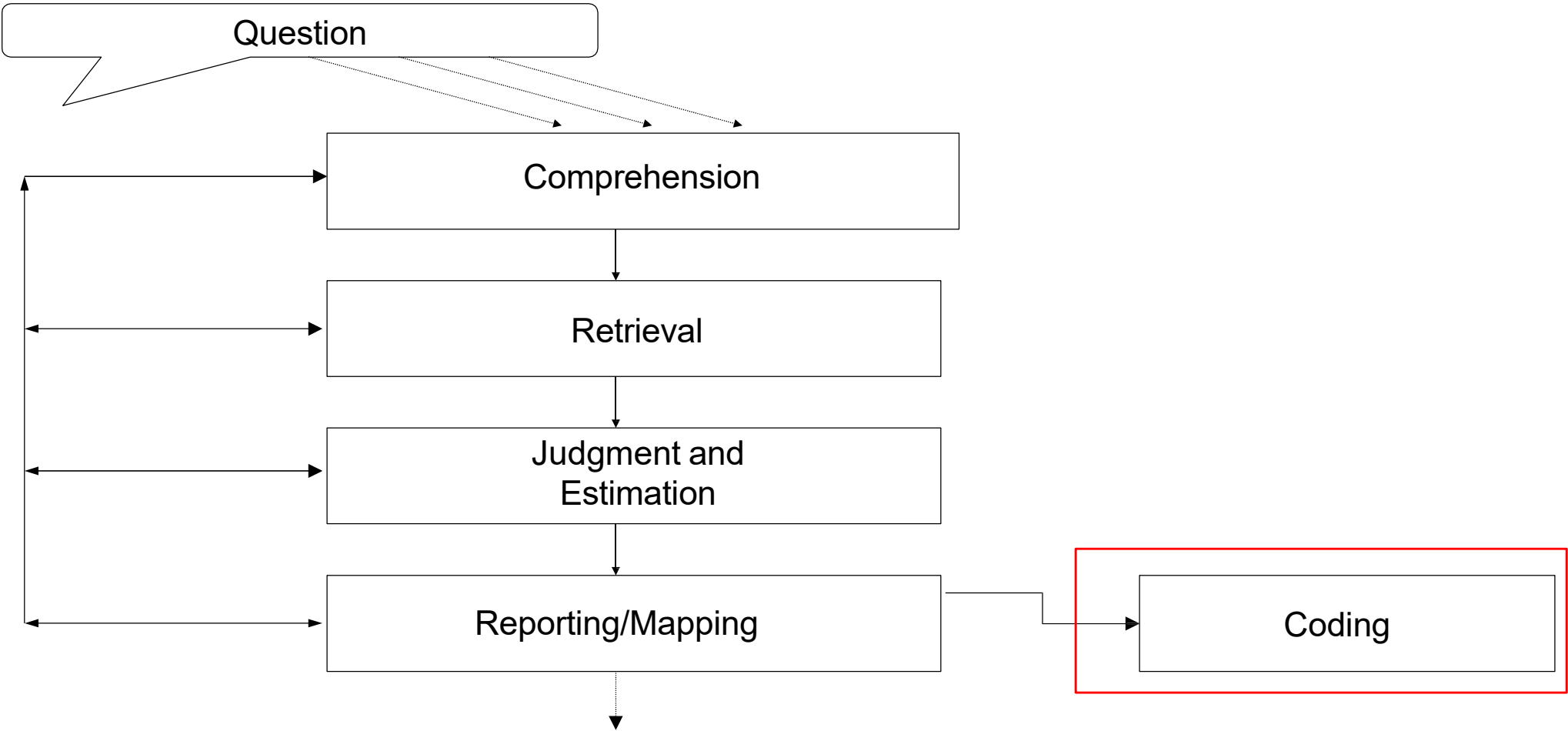
Response Mapping/Selection (2)

- Sensitive questions
 - Motivated misreporting (Tourangeau & Yan, 2007): *Rs* underreport sensitive (embarrassing, stigmatized, illegal) behaviors
 - Especially when *Q* asked by an *I'wer*
 - More sensitive information disclosed in self-administered modes
 - Are *R*'s lying to the *I'wer*?

	Time Frame		
	Past Month	Past Year	Lifetime
Cocaine	1.74	2.84	1.81
Marijuana	1.66	1.61	1.48

Ratios of Estimated Prevalence under Self-administration (ACASI) and
Interviewer-administration (CAPI)

Source: Tourangeau & Smith, 1996

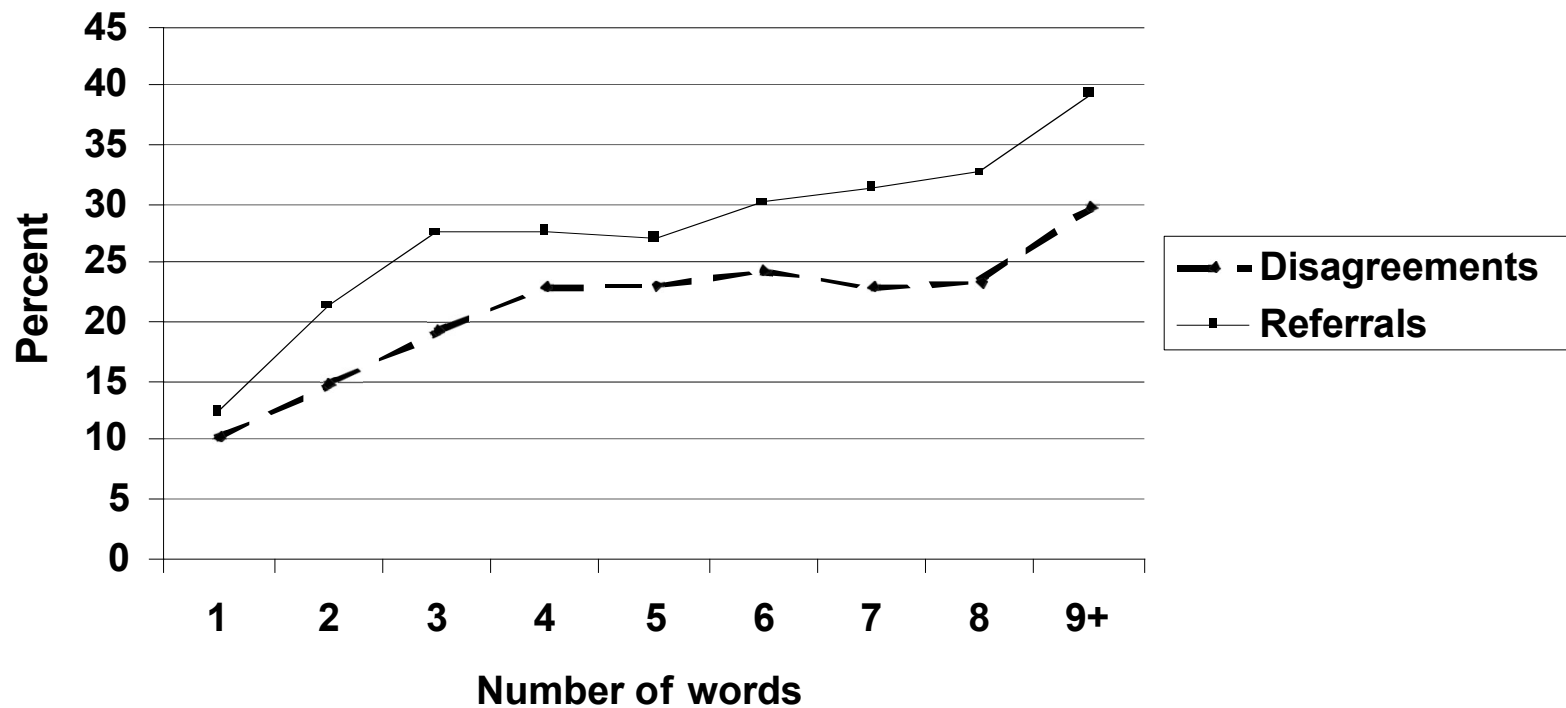


Coding:

Classifying open responses

- Questions that do not provide specific response options, i.e., “open questions,” lead to answers that cannot be tallied in raw form
 - e.g., “What are the duties of your main job?”
- By coding open answers, i.e., assigning similar answers to the same category, they can be tallied like closed answers
- Coding involves human decisions which inevitably introduce error
- Conrad, Couper & Sakshaug (2016) reported that longer descriptions (i.e., more words) of occupations led to lower inter-coder agreement than shorter descriptions
 - while high agreement doesn’t mean both parties are right, low agreement means at least one is wrong
 - surprising because extra info might resolve ambiguity

Length of description and disagreement/referral percentages



Source: Conrad, Couper & Sakshaug, 2016

Measurement Error: Cognitive and Social Origins

- Survey statistics and psychometric views of measurement error
- Cognitive views of error
- Combining the two approaches

Measurement Error

- Individual survey responses may depart from the *true value* –
 - This difference is measurement error
 - Although the objective truth may be unknowable
- Disciplinary differences in how think about measurement error:
 - Statistical: $y_{jt} = X_j + \rho_{jt}$
 - Psychometric:
 - Reliability – same results for > 1 test?
 - Validity – are the results accurate
 - Cognitive: discrepancy due to misunderstanding, forgetting, recency, etc.
- but all share concern that inaccurate responses undermine survey estimates
- and struggle with how to know the “truth”

What is True Value in each case?

Can it be determined?

1. Self-reported travel underreports travel according to GPS on smartphone
2. Pre-election poll (survey) underestimates a candidate's electoral victory
3. Respondents overreport religious attendance when directly asked compared to time use interview in which asked to reconstruct all activities yesterday
4. More reporting of cocaine use when question self-administered than when asked by an interviewer
5. Higher correlation between previous specific question (e.g., about marriage) and subsequent general question (e.g., about life happiness) when specific precedes general question than vice versa
6. Tendency to select earlier than later response options when textual question is self-administered as in a web survey